



Inspiring Excellence

CSE461

Section : 8

Group : 3

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Robot Project Proposal

Member Details:

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Project Idea 1

1. Project Title: Smart Autonomous Hospital Delivery & Safety Monitor Robot

2. Purpose

Objective: To automate the distribution of lightweight clinical materials (including medications, diagnostic lab samples, and charts) within a designated facility structure while verifying cargo climate safety, monitoring for fire hazards, and tracking structural barriers on the route.

Scope: The system operates within a controlled indoor track architecture replicating a hospital interior grid (comprising a Pharmacy, Laboratory, Nurse Station, and individual Wards) using coordinate tag markers for local routing verification.

Significance: Minimizes non-clinical walking workloads for nursing personnel, keeps critical supplies secure using target-locked structural container gates, actively monitors payload storage environment context, and reduces human error during hospital point-to-point drop-offs.

3. Components

- **Microcontroller:** Arduino Uno R3
- **Sensors:** IR Sensor Array, Ultrasonic Proximity Sensor, Read/Write Module, DHT11 Temperature Sensor, IR Flame Sensor Module.
- **Actuators:** 2x 5V DC Geared Motors, SG90 Micro Servo Motor, 5V Active Buzzer, Common Cathode RGB Status LED.
- **Body/Chassis:** 2-Wheel Drive (2WD) Dual-Layer Smart Acrylic Chassis featuring an Omni-directional front caster wheel.
- **Additional Components:** PCF8574 I2C I/O Expansion Module Breakout, 16x2 LCD Screen with I2C Backpack Interface, L298N Dual H-Bridge Motor Driver Module, 2x 18650 Rechargeable 3.7V Li-ion Batteries (configured in series for 7.4V layout), Dual-slot 18650 Battery Housing, 5x Tactile Pushbuttons with plastic extensions, Breadboard, structural standoffs, and stranded connection jumpers.

4. Cost Breakdown (Approx.)

Components	Unit Cost	Quantity	Total
Arduino Uno R3	950	1	950
2WD Acrylic Car Chassis Set (inc. DC Motors/Caster)	570	1	570

L298N H-Bridge Motor Driver	185	1	185
MFRC522 RFID Module with Key Fob/Card Tags	190	1	190
16x2 LCD Display with integrated I2C Backpack	255	1	255
PCF8574 I2C I/O Expansion Module Breakout	160	1	160
TCRT5000L 5-Channel IR Line Tracking Array	310	1	310
HC-SR04 Ultrasonic Sensor Module	130	1	130
DHT11 Temperature Sensor Module	120	1	120
IR Flame Sensor	90	1	90
SG90 Micro Servo Motor	130	1	130
18650 Rechargeable Li-ion Batteries	260	2	520
Misc (Buzzer, Pushbuttons, LEDs, Chassis Standoffs, Wires)	250	1	250
Total			3860

5. Functionality Breakdown

- **Functionality 1: Autonomous Grid Navigation & Intersection Coordinates Routing**

- **Overview:** Navigates floor tracks autonomously via a high-precision 5-channel infrared differential detection system and executes localized directional adjustments at crossroads by scanning under-floor coordinate identifier blocks.
- **Working Procedure:** The TCRT5000L 5-channel IR array reads floor contrast configurations with enhanced sensitivity, feeding precise binary track positions directly to the Arduino Uno's internal tracking loop to balance left/right PWM duty cycles on the L298N driver. Having 5 channels allows the robot to handle sharp curves and alignment much more smoothly. When encountering an intersection cross-line, the chassis halts momentarily while the bottom-facing MFRC522 reads an embedded RFID tracking sticker. The logic cross-references this read UID against a coordinate string matrix populated by the user's initial pushbutton choices on the PCF8574 module. If a path variation is needed to reach the target ward, one motor reverses to execute a precise 90-degree pivot until the tracking line is re-acquired by the center IR sensors.
- **Functionality 2: Intelligent Collision Avoidance & Safety-Lock Payload Guarding**
 - **Overview:** Scans for structural line hazards or pedestrians dynamically and implements a secure mechanical door control assembly on the medicine safe box that operates only at the matching destination.
 - **Working Procedure:** The front-mounted HC-SR04 ultrasonic sensor transmits acoustic pulses continuously, computing runtime echo gaps via native timing captures. If an obstacle distance scales under 20 cm, the core loop halts motor execution, keeping the robot stationary while cycling a warning message across the I2C LCD screen until the path clears. The cargo compartment door remains physically closed by the SG90 servo motor arm (positioned at 0 degrees), ensuring protection against transit theft. Once the RFID confirmation validates that the target room is reached, the servo rotates 90 degrees to lift the container latch, locking back to 0 degrees only when the target delivery confirmation button is toggled by clinical personnel.
- **Functionality 3: Multi-Hazard Environmental Monitoring & UI Telemetry Loop**
 - **Overview:** Provides ongoing background scans for temperature safe bounds inside the storage vault, screens for ambient open fires, and handles system updates over an extended input bus.
 - **Working Procedure:** An internal compartment DHT11 module checks ambient cargo conditions; if readings breach 30°C, a secondary notification flag runs on the display alongside intermittent audio alerts. Concurrently, the IR flame sensor monitors ambient space for fire flare-ups. Detecting a flame triggers a high-priority software override loop that cuts power to the motor driver instantly, locks the delivery compartment, sounds a continuous alarm tone through the

buzzer, and pulses the RGB LED red. System metrics (including target destination, active node coordinates, cargo heat readings, and low voltage safety warnings) are formatted and multiplexed over the shared A4/A5 I2C lines to update the screen layout smoothly.